Tirunelveli District

Common Quarterly Examination - September 2026

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Standard 12

PHYSICS

Time Allowed: 3.00 Hours Maximum Marks: 70

PART - I

Choose the best answer: **15×1=15**

1) An electric field $\vec{E} = 10x\hat{i}$ exists in a certain region of space. Then the potential difference $V = V_0 - V_A$ where V_0 is the potential at the origin and V_A is the potential at $x = 2m$ is
a) 10V b) -20V c) +20V d) -10V

2) Two wires of A and B with circular cross section are made up of the same material with equal length Suppose $R_A = 3R_B$, then what is the ratio of radius of wire A to that of B?
a) 3 b) $\sqrt{3}$ c) $\frac{1}{\sqrt{3}}$ d) $\frac{1}{3}$

3) A parallel plate capacitor stores a charge Q at a voltage V. Suppose the area of the parallel plate capactior and the distance between the plates are each doubled, then which is the quantity that will change?
a) capacitance b) charge c) voltage d) energy density

4) What is the flux through a cube of side 'a' if a point charge of q is at one of its corner?
a) $\frac{2q}{\epsilon_0}$ b) $\frac{q}{8\epsilon_0}$ c) $\frac{q}{\epsilon_0}$ d) $\frac{q}{2\epsilon_0} 6a^2$

5) In a large building there are 15 bulbs of 40W, 5 bulbs of 100W, 5 fans of 80W and 1 heater of 1KW are connected. The voltage of electric mains 220V. The maximum capacity of the main fuse of the building will be
a) 14A b) 8A c) 10A d) 12A

$\vec{B} = 0.2\hat{i}T$

6) An electric bulb is rated 60W, 220V. Then resistance of the filament is,
a) 870Ω b) 780Ω c) 708Ω d) 807Ω

7) Franhofer lines are an example of _____ spectrum.
a) line emission b) line absorption c) band emission d) band absorption

8) The dimension of $\frac{1}{\mu_0\epsilon_0}$ is
a) $[LT^{-1}]$ b) $[L^2T^{-2}]$ c) $[L^{-1}T]$ d) $[L^{-2}T^2]$

9) The flux linked with a coil at any instant t is given by $\phi_B = 10t^2-50t+250$. The induced emf at $t = 3s$ is
a) -190V b) -10V c) 10V d) 190V

10) A step down transformer reduces the supply voltage from 220V to 11V and increases the current from 6A to 100A. Then its efficiency is
a) 1.2 b) 0.12 c) 0.83 d) 0.9

11) In Which of the following devices, the eddy current effect is not used.
a) electric heater b) induction furnace
c) magnetic braking in train d) electromagnet

12) The potential energy of magnetic dipole whose dipole moment is $\vec{P}_m = (0.5\hat{i} + 0.4\hat{j})Am^2$ kept in uniform magnetic field $\vec{B} = 0.2\hat{i}T$ is
a) -0.1J b) -0.8J c) 0.1J d) 0.8J

13) A thin insulated wire forms a plane spiral of N = 100 tight turns carrying a current I = 8mA. The radius of inside and outside turns are a = 50mm and b=100mm respectively. The magnetic induction at the centre of the spiral is
a) 5μT b) 7μT c) 8μT d) 10μT

14) Stars twinkle due to
a) reflection b) total internal reflection c) refraction d) polarisation

15) A rod of length 10 cm lies along the principal axis of a concave mirror of focal length 10 cm in such a way that its end closer to the pole is 20 cm away from the mirror. The length of the image is,
a) 2.5 cm b) 5 cm c) 10 cm d) 15 cm

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PART - II

Answer any six of the following questions. Question No. 20 is compulsory: **6×2=12**

16) What is critical angles and total internal reflection?
17) Define: Temperature coefficient of resistance.
18) State: Biot Savart law.
19) Define: Electric dipole. Give the expression for the magnitude of its electric dipole moment and the direction.
20) Compute the speed of the electromagnetic wave in a medium if the amplitude of electric and magnetic fields are 3×10^{4} NC^{-1} and 2×10^{-4}T respectively.
21) What is displacement current?
22) How will you define RMS value of an alternating current?
23) Compute the magnitude of magnetic field of a long straight wire carrying a current of 1A at a distance of 1m from it. Compare it with Earth's magnetic field.
24) Define: Q factor.

PART - III

Answer any six of the followings. Question No. 28 is compulsory: **6×3=18**

25) Obtain Gauss law from couloumb's law.
26) Write short notes on (a) Microwave (b) X ray
27) Explain the equivalent resistance of a series resistor network.
28) Calculate the electric flux through a rectangle of sides 5 cm and 10 cm kept in the region of a uniform electric field 100 NC^{-1}. The angle θ is 60°. If θ becomes zero, What is the electric flux?
29) Derive the relation between f and R for a spherical mirror.
30) Give the uses of Focault current. (any three)
31) Light travelling through transparent oil enters into glass of refractive index 1.5. If the refractive index of glass with respect to the oil is 1.25. What is the refractive index of the oil?
32) Give an account of magnetic Lorentz force.
33) What are the advantages of AC over DC?

PART - IV

Answer all the questions: **5×5=25**

34) Explain in detail the construction and working of a Vande-Graff Generator.

(OR)

Obtain the lens maker's formula and mention its significance.

35) Obtain the condition for bridge balance in Wheatstone's bridge.

(OR)

Calculate the magnetic field at a point on the axial line of a bar magnet.

36) Derive the expression for phase angle between the applied voltage and current in a series RLC circuit.

(OR)

Describe Fizeau's method to determine the speed of light.

37) a) Obtain the relation between current and drift velocity.
b) A copper wire of 10^{-6} m^2 area of cross section carries a current of 2A. If the number of free electrons per cubic metre in the wire is 8×10^{28}. Calculate the drift velocity of electrons.

(OR)

a) Obtain the expression for capacitance of a parallel plate capacitor.
b) A parallel plate capacitor filled with mica having $\epsilon_r = 5$ is connected to a 10V battery. The area of each parallel plate is $6cm^2$ and seperation distance is 6mm. Find the capacitance value.

38) Show Mathematically that the rotation of a coil in a magnetic field over one rotation induces an alternating emf of one cycle.

(OR)

Explain the types of Emission spectra.

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Answer Key

Std : 12 Physics

PART - I

I) Choose the best Answers:

1) c) +20V

solution: $dV = -\vec{E}\cdot d\vec{x}$

$\int_{V_0}^{V_A} dV = -\int_0^2 10x\,dx = -10\left[\frac{x^2}{2}\right]_0^2$

$= -\frac{10}{2}[2^2 - 0^2] = -5\times 4$

$= -20$

Since, $V = V_0 - V_A$ (Given).

$= -(-20V) = +20V$ //

2) c) $\frac{1}{\sqrt{3}}$

Hint: They have same material (ρ) & same length (l)

Solution: $R = \frac{\rho l}{A} = \frac{\rho l}{\pi r^2}$

$\therefore \frac{R_A}{R_B} = \left(\frac{r_B}{r_A}\right)^2$ [$\because R_A = 3R_B$ Given]

$3 = \left(\frac{r_B}{r_A}\right)^2$

Taking square root $\frac{r_B}{r_A} = \sqrt{3}$.

Inverting the ratio, $\frac{r_A}{r_B} = \frac{1}{\sqrt{3}}$ //

3) d) Energy density

4) b) $\frac{q}{8\varepsilon_0}$ Explanation: According to Gauss's law, if a point charge is kept at the corner of a cube, it is shared equally among 8 identical cubes.

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5) d) 12A Explanation: Total Power = (15 × 40W) + (5 × 100W) + (5 × 80W)

$P = VI, \Rightarrow I = \frac{P}{V}$

$= \frac{2500}{220} \approx 11.36A$

6) d) 807 Ω Explanation: $P = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P}$

$R = \frac{200^2}{60} = \frac{200 \times 200}{60} \approx 806.6\Omega$

$R = 807\Omega$

7) b) line absorption.

8) b) L^2T^{-2}

9) b) −10V Exp: $e = -\frac{d\phi}{dt}$; $\phi_B = 10t^2 - 50t + 200$ ... (1)

$\therefore \frac{d\phi}{dt} = 10 \times 2t - 50(1)$

$= 20t - 50$

(1) $\Rightarrow e = -(20t - 50)$ [$\because t = 3s$]

$e = -[(20 \times 3) - 50]$

$= -[60 - 50] = -10V$

$e = -10V$

10) c) 0.83 Exp: $\eta = \frac{V_s I_s}{V_P I_P} = \frac{220V \times 100A}{220V \times 6A} \approx 0.83$

11) a) electric heater Exp: electric heater generates heat directly through Joule's law of heating $H = I^2Rt$

12) a) −0.1 J Exp: $U = -\vec{P_m} \cdot \vec{B} = -(0.5\hat{i} + 0.4\hat{j}) \cdot 0.2\hat{i}$

$= -0.5 \times 0.2$

$U = -0.1\ J$

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13) b) 7 μT

Exp: $B = \frac{\mu_0 N I}{2(y-x)} \int_a^b \frac{dr}{r}$

$= \frac{\mu_0 N I}{2(y-x)} \ln\left(\frac{y}{x}\right)$

Given: $N = 100$, $I = 8\text{mA} = 8\times10^{-3}\text{A}$

$x = 50\text{mm} = 0.05\text{m}$, $y = 100\text{mm}$

$y - x = 0.05\text{m}$.

$\ln \frac{y}{x} = \ln 2 = 0.693$

$B = \frac{4\pi\times10^{-7}\times 8\times10^{-3}\times 100 \times 0.693}{2\times 0.05}$

$= \frac{4\pi\times10^{-7}\times 0.8\times 0.693}{0.1}$

$= 6.97\times10^{-6}\text{T}$

$B = 7\,\mu\text{T}$ //

14) c) Refraction.

15) b) 5 cm

Exp: $f = 10\text{cm}$, $l = 10\text{cm}$

Near end, $u_1 = -20\text{cm}$

$\frac{1}{v_1} = \frac{1}{f} - \frac{1}{u_1} = \frac{1}{-10} - \frac{1}{-20} = \frac{1}{-10} + \frac{1}{20}$

$= \frac{20-10}{200} = \frac{10}{200} = \frac{1}{20}$

$v_1 = 20\text{cm}$

Far end, $u_2 = -(20+10) = -30\text{cm}$

$\frac{1}{v_2} = \frac{1}{-10} - \frac{1}{-30} = \frac{1}{-10} + \frac{1}{30} = \frac{30-10}{300}$

$= \frac{20}{300}$

$v_2 = \frac{30}{2} = 15\text{cm}$ //

$|v_1 - v_2| = |-20 - (-15)| = |-20 + 15|$

$= 5\text{cm}$ //

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Part - II

20) Given Data:

$E = 3\times10^{4}\ NC^{-1}$, $B = 2\times10^{-4}\ T.$, $v = ?$

Soln: $v = \frac{E}{B} = \frac{3\times10^{4}}{2\times10^{-4}} = 1.5\times10^{8}\ ms^{-1}$ //

23) Given: $I = 1A$, $r = 1m$, $B = ?$

$B = \frac{\mu_0 I}{2\pi r} = \frac{4\pi\times10^{-7}\times1}{2\times\pi\times1} = 2\times10^{-7}\ T$ //

Part - III

28) Soln: $\phi_E = E \cdot A\cos\theta$ $E = 100\ NC^{-1}$, $A = 5\times10\ 250cm^2$

(i) $\phi_E = 100\times50\times\cos60^\circ$ (ஃ $\theta = 60^\circ$)

$= \frac{250}{500}\times\frac{1}{2}\ cm^2 = 0.25\ Nm^2C^{-1}$

(ii) $\phi_E = 100\times50\ cm^2\times\cos0^\circ$ [∵ if $\theta = 0^\circ$]

$= 500\times10^{-4}m^2 = 0.5\ Nm^2C^{-1}$

31) Soln: $\mu_g = \frac{\mu_g}{\mu_0} \Rightarrow \mu_0 = \frac{\mu_g}{\mu} = \frac{1.5}{1.25} = 1.2.$

$\boxed{\mu_0 = 1.2}$

Part - IV

37) b) $I = nAev_d$

$v_d = \frac{I}{nAe} = \frac{2}{8\times10^{28}\times10^{-6}\times1.6\times10^{-19}} = \frac{2}{12.8\times10^{3}} = \frac{2}{12800}$

$= 1.5625\times10^{-4}\ ms^{-1}$

$\boxed{v_d \approx 1.56\times10^{-4}\ ms^{-1}}$ [OR]

b) Soln: $C = \frac{\varepsilon_0\varepsilon_r A}{d} = \frac{8.854\times10^{-12}\times5\times6\times10^{-4}}{6\times10^{-3}} = 4.43\times10^{-12}\ F$

$C = 4.43\times10^{-12} f$

$\boxed{C = 4.43pf}$

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